2.4 Spatial and geometric representations

Think back to the first time you heard the term 'digital twin'. Chances are, you pictured a rich 3D visual model that closely represents reality. There is a good reason for this: many modern digital twins incorporate 3D models of the physical system they mirror, for a number of reasons

- A 3D model provides an immersive spatial context that allows people to visualize a complex system as it actually exists in the physical world, aiding in their understanding of the size, shape, location, and relationship of physical objects.
- Overlaying a 3D model of a built asset over that of the design provides a powerful mechanism to see where the built asset differs from the design view.
- The models can be used to simulate physical phenomena in the real world, such as the flow of air over a newly designed vehicle.
- They are well suited to the contextualization of data received about the physical environment via sensors, allowing it to be spatially located, which then aids in the consumption and understanding of the data.

In addition to the immersive realism experience offered by 3D visualizations, 2D geometric representations that show layouts and measurements are no less important when capturing a digital representation of reality. Just like engineering documents and drawings, 2D geometric representations of physical spaces often employ domain-specific visual abstractions that clearly convey information about the physical world without the complexity that a 3D spatially accurate model may bring.

2.4.1 2D geometric models

Long before computer graphics enabled photorealistic 3D models, maps, floorplans, and blueprints allowed engineers, architects, planners, and operators to understand, design, and manage complex spatial relationships in the real world. In digital twin implementations, these traditional formats retain their importance for the reasons that they are information dense,